

SANNIKOVA, Russian Federation

WMO: 215350

Lat: 74.667N

Lon: 138.900E

Elev: 15

StdP: 101.15

Time Zone: 10.00 (E10)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.2	-38.8	-42.6	0.1	-39.3	-41.3	0.1	-38.1	19.0	-20.6	16.4	-24.0	3.1	90	1.179

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.0	8.9	7.3	7.1	6.0	5.9	5.0	7.5	8.6	6.1	6.9	5.0	5.8	5.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.6	6.1	8.3	5.4	5.6	6.7	4.2	5.1	5.5	23.9	8.7	21.0	7.0	18.7	5.9	12.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.6	13.0	11.6	DB	-42.2	12.1	1.3	3.2	-43.1	14.4	-43.9	16.3	-44.6	18.1	-45.6	20.5	(4)
(5)			WB	-42.2	9.6	1.3	1.2	-43.1	10.5	-43.9	11.2	-44.6	11.9	-45.6	12.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-13.6	-29.9	-30.5	-27.2	-18.6	-7.8	0.6	2.5	2.6	-0.1	-8.5	-20.6	-27.2	(6)	
	DBStd	13.58	4.92	5.66	5.73	5.60	4.16	2.53	1.93	2.32	2.89	5.87	6.14	5.21	(7)	
	HDD10.0	8627	1238	1133	1154	857	552	283	232	229	303	575	918	1153	(8)	
	HDD18.3	11669	1496	1366	1413	1107	810	533	490	488	553	833	1168	1412	(9)	
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	Wind	WSAvg	5.3	4.7	4.8	5.1	5.7	6.1	5.5	5.4	5.4	5.6	5.5	4.9	4.7	(14)
	Precipitation	PrecAvg	276	13	13	12	11	16	24	37	44	37	33	22	14	(15)
(15) Precipitation	PrecMax	381	31	41	48	40	44	62	63	89	69	76	44	49	(16)	
	PrecMin	166	2	1	1	1	3	5	15	16	11	10	6	2	(17)	
	PrecStd	52	7	11	9	9	10	13	12	19	14	14	10	11	(18)	
	PrecStd	52	7	11	9	9	10	13	12	19	14	14	10	11	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-14.2	-12.9	-6.8	-4.4	1.7	11.8	12.5	11.2	6.3	1.4	-3.7	-12.8	(19)	
		MCWB	-14.7	-13.3	-7.4	-5.0	0.6	9.2	10.2	9.5	5.1	0.7	-4.4	-13.3	(20)	
	2%	DB	-17.7	-16.2	-9.4	-6.7	0.5	8.0	8.6	8.5	5.1	0.6	-6.1	-15.6	(21)	
		MCWB	-17.9	-16.4	-9.8	-7.3	-0.2	6.1	7.0	7.3	4.1	-0.1	-6.7	-15.8	(22)	
	5%	DB	-20.0	-19.6	-15.2	-8.4	-0.3	5.4	6.7	7.1	4.3	-0.2	-8.9	-18.2	(23)	
		MCWB	-20.2	-19.9	-15.4	-8.9	-0.8	4.1	5.5	6.2	3.5	-0.7	-9.3	-18.5	(24)	
	10%	DB	-22.2	-22.3	-19.3	-10.1	-1.3	3.5	5.4	6.1	3.6	-1.2	-11.6	-19.8	(25)	
		MCWB	-22.5	-22.5	-19.6	-10.6	-2.0	2.6	4.4	5.4	2.9	-1.9	-11.9	-20.0	(26)	
	0.4%	WB	-14.5	-13.3	-7.4	-4.8	0.7	9.2	9.7	9.6	5.3	1.0	-4.3	-13.3	(27)	
		MCDB	-14.3	-13.1	-7.0	-4.5	1.2	11.5	11.6	11.0	6.2	1.2	-3.8	-12.9	(28)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-17.8	-16.4	-9.9	-7.2	0.0	6.2	7.1	7.5	4.3	0.1	-6.6	-15.9	(29)	
		MCDB	-17.6	-16.2	-9.4	-6.7	0.3	7.9	8.2	8.2	4.9	0.5	-6.1	-15.6	(30)	
	5%	WB	-20.3	-19.8	-15.4	-8.9	-0.8	4.2	5.5	6.3	3.6	-0.7	-9.3	-18.4	(31)	
		MCDB	-20.2	-19.6	-15.2	-8.4	-0.3	5.4	6.5	6.8	4.1	-0.3	-8.9	-18.2	(32)	
	10%	WB	-22.4	-22.4	-19.5	-10.6	-2.0	2.6	4.5	5.4	2.9	-1.8	-11.9	-20.0	(33)	
		MCDB	-22.2	-22.3	-19.3	-10.1	-1.3	3.4	5.3	6.0	3.5	-1.2	-11.5	-19.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.5	6.1	7.3	4.4	3.0	3.5	3.0	2.8	4.3	5.2	5.1	(35)	
		MCDBR	7.6	8.3	9.8	8.6	4.0	6.0	6.6	5.1	2.8	2.8	6.4	7.4	(36)	
	5% WB	MCWBR	7.5	8.3	9.5	8.3	4.0	5.1	5.4	4.3	2.6	2.8	6.3	7.3	(37)	
		MCDBR	7.6	8.3	9.7	8.5	4.1	6.0	6.2	4.6	2.7	2.7	6.6	7.4	(38)	
	5% WB	MCWBR	7.6	8.2	9.6	8.3	4.1	5.1	5.3	4.0	2.6	2.8	6.5	7.3	(39)	
		MCWBR	7.6	8.2	9.6	8.3	4.1	5.1	5.3	4.0	2.6	2.8	6.5	7.3	(39)	
(40) Clear-Sky Solar Irradiance	taub	N/A	0.169	0.268	0.301	0.331	0.322	0.354	0.344	0.288	N/A	N/A	N/A	N/A	(40)	
	taud	N/A	1.878	1.987	2.012	2.013	2.206	2.281	2.376	2.361	N/A	N/A	N/A	N/A	(41)	
	Ebn at Noon	0	299	613	767	803	839	788	738	641	N/A	0	0	0	(42)	
	Edh at Noon	0	19	70	107	127	109	98	77	51	N/A	0	0	0	(43)	
(44) All-Sky Solar Radiation	RadAvg	0.00	0.10	1.35	3.92	6.15	6.59	5.16	2.72	1.17	0.22	0.00	0.00	0.00	(44)	
	RadStd	0.00	0.01	0.07	0.17	0.22	0.34	0.28	0.33	0.10	0.02	0.00	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page